SmartHerbalNet: IoT-Assisted Environmental Data Collection and Leaf Disease Prediction Through Transfer Learning Approach

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Abstract

SmartHerbalNet uses AI and IoT to detect diseases in herbal leaves and monitor environmental conditions in real-time. Transfer learning classifies leaf health from a dataset of 10 herbal plant categories, while IoT sensors track soil and climate data. Farmers are alerted if conditions deviate from optimal, supporting healthier crops and higher production.

Introduction

Herbal farming is essential for medicine and nutrition, but leaf diseases and unfavorable environmental conditions can reduce yield and quality. SmartHerbalNet leverages AI and IoT to detect leaf diseases from images using transfer learning and monitor environmental conditions in real-time. The system alerts farmers when parameters deviate from optimal ranges, enabling timely interventions to maintain healthy plants and improve productivity.

Motivation

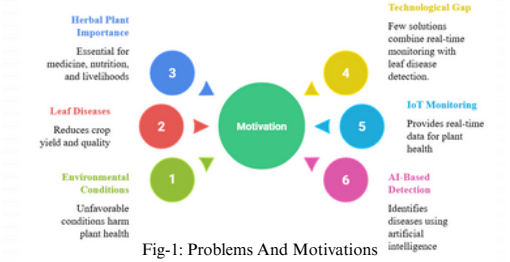


Fig-1: Problems And Motivations

Research Questions

- How can AI be used to accurately detect diseases in herbal plant leaves?
- What are the key environmental factors affecting the health of herbal plants in real-time?
- How can integrating AI and IoT support farmers in improving yield and preventing losses?

Objectives

- To collect and prepare data from herbal leaf images and IoT-based environmental sensors for AI model training.
- To develop a transfer learning model to accurately detect healthy and diseased herbal leaves.
- To integrate AI-based leaf disease detection with real-time IoT environmental monitoring to alert farmers about unfavorable condition.
- To evaluate the system's performance in terms of disease detection, environmental monitoring, and effectiveness of farmer alerts.

Literature Review

Paper	Contribution	Limitation
Sambana et al(2025)[1]	Applied transfer learning with YOLOv7/YOLOv8 for plant disease detection.	Focused only on common crop diseases. No exploration of herbal plant diseases.
Preotee et al.(2024)[2]	Developed a Bangladeshi leaf disease dataset and applied transfer learning with XAI.	Excludes herbal plants, and lacks real-time environmental data monitoring.
Saiful Islam(2023) [3]	2,029 high quality images of 10 medicinal plants and open-access for research and education.	Lacking of real-time environmental data. No disease classification.
Shahab(2025) [4]	Real-time monitoring of environmental data and integrated AI-driven mobile app for crop management	Only tested on rice. No long-term monitoring and no specific ML model used.

Methodology

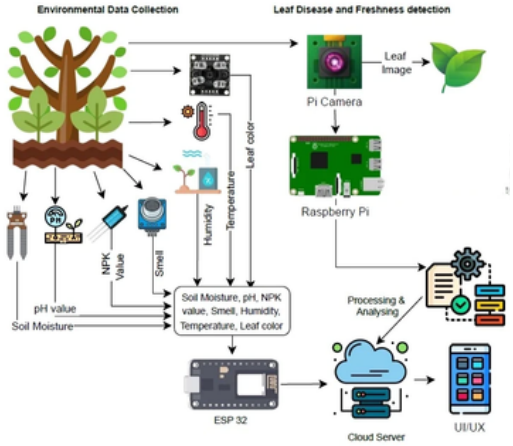


Fig-2: System architecture

Expected Outcome

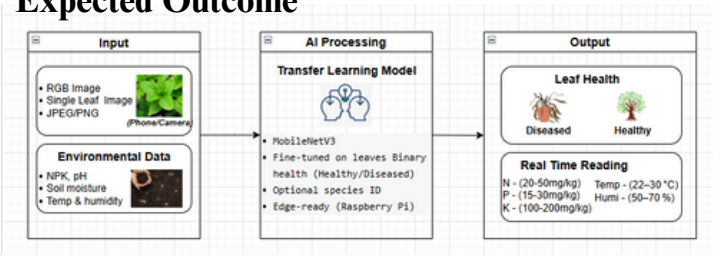


Fig-3: System Workflow

Conclusion

SmartHerbalNet combines IoT and transfer learning-based AI to support smart herbal farming by enabling real-time environmental monitoring and accurate leaf disease detection. The system alerts farmers when soil or climate conditions deviate from optimal, helping prevent crop losses and maintain plant health. By providing actionable insights, SmartHerbalNet improves herbal crop yield, quality, and overall farm management efficiency.

Future Work



Fig-4: Future Work

Reference

[1] Sambana, B., Nnadi, H. S., Wajid, M. A., Nwosu, O. F., Camacho-Zuñiga, C., DozieAjuzie, H., & Onyema, E. M. (2025). An efficient plant disease detection using transfer learning approach. Scientific Reports, 15, Article 19082. https://doi.org/10.1038/s41598-025-02271-w

[2] Preotee, F. F., Sarker, S., Refat, S. R., Muhammad, T., & Islam, S. (2024, December). An approach towards identifying Bangladeshi leaf diseases through transfer learning and XAI. In 2024 27th International Conference on Computer and Information Technology (ICCIT) (pp. 1–6). IEEE. https://doi.org/10.1109/ICCIT64611.2024.11022335

[3] Islam, S., Ahmed, M. R., Islam, S., Rishad, M. M. A., Ahmed, S., Utshow, T. R., & Siam, M. I. (2023). BDMediLeaves: A leaf images dataset for Bangladeshi medicinal plants identification. Data in Brief, 50, 109488. https://doi.org/10.1016/j.dib.2023.109488

[4] Shahab, H., Naeem, M., Iqbal, M., Aqeel, M., & Ullah, S. S. (2025). IoT-driven smart agricultural technology for real-time soil and crop optimization. Smart Agricultural Technology, 10, 100847. https://doi.org/10.1016/j.atech.2025.100847